

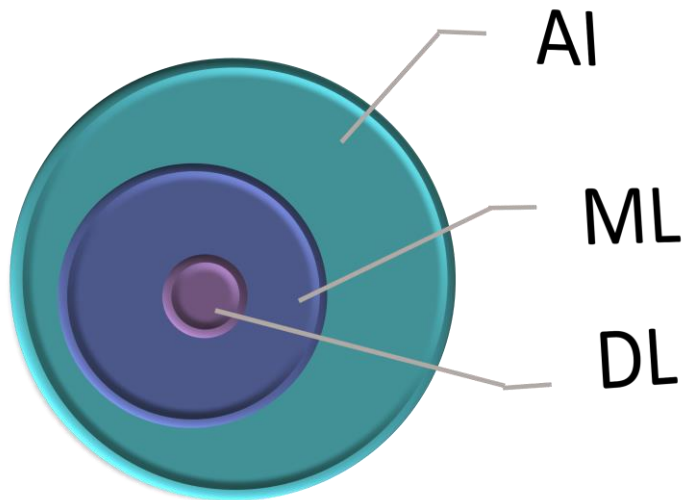
بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

كلية الدراسات العليا – جامعة النيلين
قسم علوم الحاسوب – مسار الذكاء الاصطناعي
Subject : argentic AI Task (1)

إعداد :

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1 – Draw a concept map illustrating the relationship among AI ,ML,DL



AI : Artificial intelligence (AI) is an area of computer science that emphasizes the creation of intelligent machines that work and react like humans.

ML: Machine Learning at its most basic is the practice of using algorithms to parse data, learn from it, and then make a determination or prediction about something in the world.

DL: Deep Learning has enabled many practical applications of Machine Learning and by extension the overall field of AI. Deep Learning breaks down tasks in ways that makes all kinds of machine assists seem possible .

2- Write two –pages report on one real-world application of deep learning

Claude AI

What is Claude AI?

Claude AI is a conversational assistant developed by Anthropic that helps you write, analyze information, and solve problems through natural dialogue. You interact with it by typing prompts or questions, and it responds in clear, structured language.

How does Claude AI work?

Claude AI works by analyzing your prompts and generating responses using advanced language models trained to understand and produce humanlike text. While the technology behind it is complex, the experience is simple: You ask questions or give instructions, and Claude responds in a clear, conversational way.

Claude AI vs. ChatGPT: What’s the difference?

Training method

Context window

Reasoning

Coding

Processing speed

How did Claude get its name? The name Claude is in reference to mathematician Claude Shannon, who was known as the “father of the Information Age.”

What is Claude Sonnet?

Claude Sonnet is one of Anthropic’s latest Claude AI models, known for improved speed, reasoning, and tool use

How to use Claude AI?

1. Writing and editing

2. Reasoning and research
3. Coding help
4. Summarizing and analyzing files
5. Visual understanding
6. Provide context

Advantages of using Claude AI

Prioritizes safe, reliable use

Handles long and complex content

Adapts to your needs over time Supports multistep and complex tasks

Delivers fast, efficient responses

What is Claude Opus?

Claude Opus is Anthropic's premium Claude AI model designed for enterprise and large-scale analytical work.

Is Claude AI free? Yes

How do I get access to Claude AI?

To start using Claude, simply sign up for an account on Anthropic's website. After you log in, you can use Claude directly in the browser and choose between the free version or a paid plan for more advanced features.

Disadvantages of using Claude AI:

May generate inaccurate information

Requires web search to be enabled for current information

Has usage limits and costs

Lacks built-in image generation

Offers limited plug-and-play customization

3- Research a recent Large Language Model (LLM) and summarize its architecture and applications

Deep Seek V3

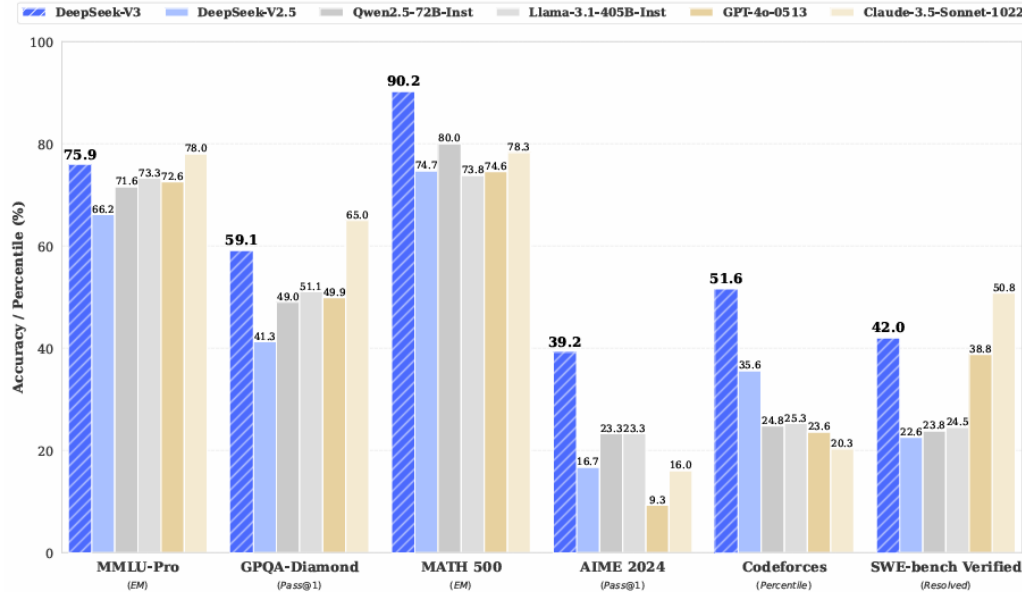


Figure 1 | Benchmark performance of DeepSeek-V3 and its counterparts.

1-Introduction :

In recent years, Large Language Models (LLMs) have been undergoing rapid iteration and evolution (Anthropic, 2024; Google, 2024; OpenAI, 2024a), progressively diminishing the gap to wards Artificial General Intelligence (AGI). Beyond closed-source models, open-source models, including DeepSeek series (DeepSeek-AI, 2024a,b,c; Guo et al., 2024), LLaMA series (AI@Meta, 2024a,b; Touvron et al., 2023a,b), Qwen series (Qwen, 2023, 2024a,b), and Mistral series (Jiang et al., 2023; Mistral, 2024), are also making significant strides, endeavoring to close the gap with their closed-source counterparts. To further push the boundaries of open-source model capabilities, we scale up our models and introduce DeepSeek-V3, a large Mixture-of-Experts (MoE) model with 671B parameters, of which 37B are activated for each token.

2-Architecture

the basic architecture of DeepSeek-V3, featured by Multi-head Latent Attention (MLA) (DeepSeek-AI, 2024c) for efficient inference and DeepSeekMoE (Dai et al., 2024) for economical training. Then, we present a Multi-Token Prediction (MTP) training objective, which we have observed to enhance the overall performance on evaluation benchmarks. For other minor details not explicitly mentioned, DeepSeek-V3 adheres to the settings of DeepSeek V2 (DeepSeek-AI, 2024c).

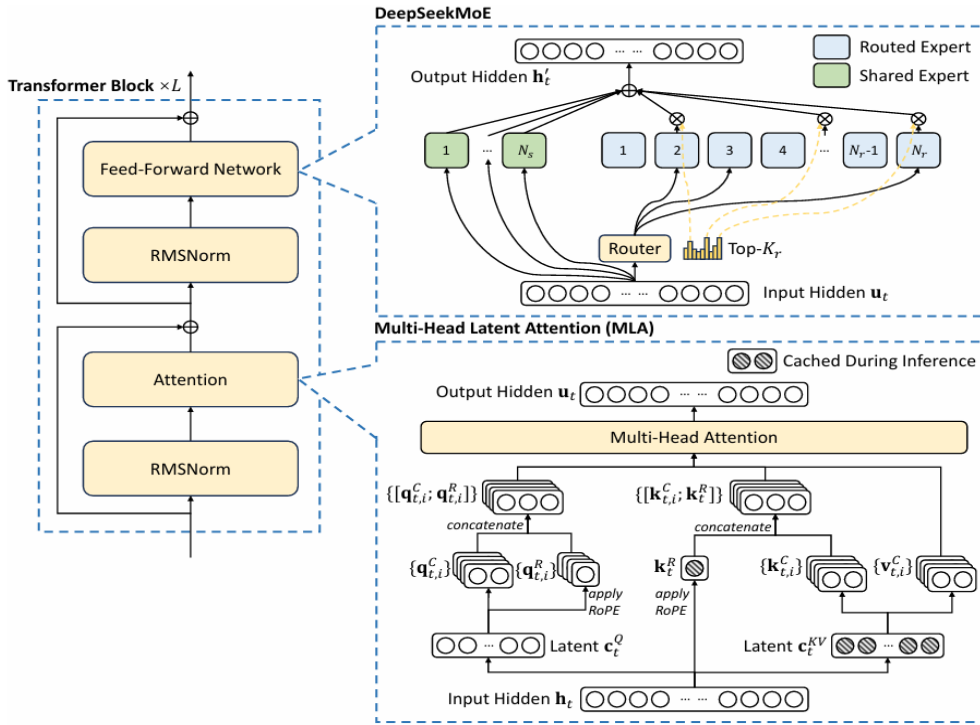


Illustration of the basic architecture of DeepSeek-V3

We adopt MLA and DeepSeekMoE for efficient inference and economical training

Multi-Head Latent Attention:

DeepSeekMoE with Auxiliary-Loss-Free Load Balancing

Complementary Sequence-Wise Auxiliary Loss

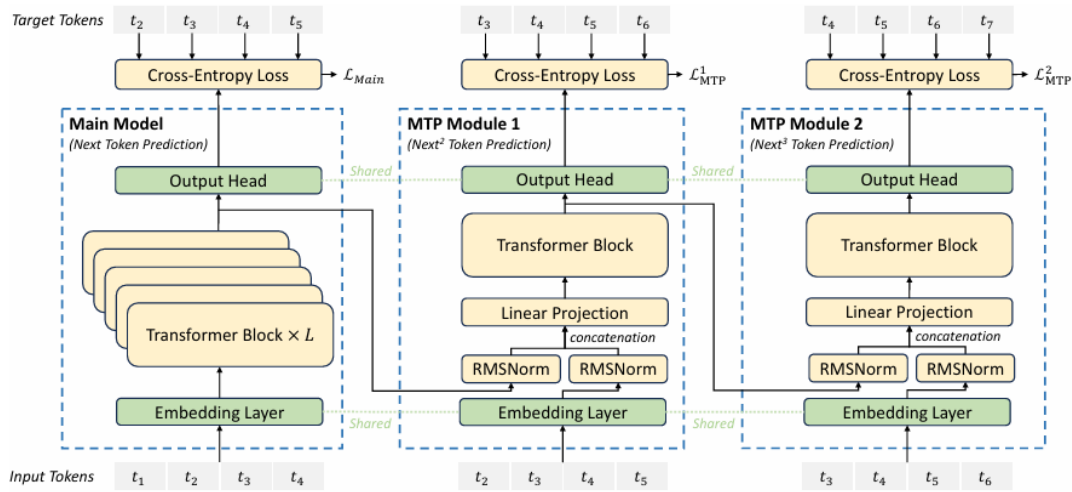


Illustration of our Multi-Token Prediction (MTP) implementation. We keep the complete causal chain for the prediction of each token at each depth.

Node-Limited Routing

NoToken-Dropping.

Multi-Token Prediction:

MTPModules.

MTPTraining Objective

MTP inInference

Infrastructures:

Compute Clusters

Training Framework

DualPipe and Computation-Communication Overlap:

Example



Method	Bubble	Parameter	Activation
1F1B	$(PP - 1)(F + B)$	$1\times$	PP
ZB1P	$(PP - 1)(F + B - 2W)$	$1\times$	PP
DualPipe (Ours)	$(\frac{PP}{2} - 1)(F + B + B - 3W)$	$2\times$	$PP + 1$

Efficient Implementation of Cross-Node All-to-All Communication:

Extremely Memory Saving with Minimal Overhead

Recomputation of RMSNorm and MLA Up-Projection

Exponential Moving Average in CPU

Shared Embedding and Output Head for Multi-Token Prediction.

FP8 Training: Mixed Precision Framework

Communication Hardware:

1-Forwarding data between the IB (InfiniBand) and NVLink domain while aggregating IB traffic destined for multiple GPUs within the same node from a single GPU.

2-Transporting data between RDMA buffers (registered GPU memory regions) and in put/output buffers.

3-Executing reduce operations for all-to-all combine.

4-Managing fine-grained memory layout during chunked data transferring to multiple experts across the IB and NVLink domain.

Application:

- 1- Coding and software developer
- 2- Math and scientific Reasoning
- 3- Enterprise and Business
- 4- Multimedia Edge AI
- 5- Content and Language
- 6- Bank and Hospitals
- 7- Companies
- 8- Universities